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TrackPilot: choosing which genomic foundation-model tracks to trust for a cell type from a few measured variants

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Abstract

Motivation: Sequence-to-function models such as AlphaGenome predict the effect of a variant on about a thousand cell-type-specific tracks. A laboratory studying one cell type must decide which of these predictions to use. Matching tracks by cell-type name is unreliable, and many cell types are not profiled at all. Laboratories often measure a few variants before a larger study, but there is no principled way to choose those variants or to turn a handful of measurements into a choice among a thousand correlated tracks.
Results: We present TrackPilot. It treats each track as a hypothesis that the assay is an unknown linear function of the track's predictions, integrates the calibration out in closed form, and starts from the tracks whose metadata match the cell type. It proposes a pilot panel of variants on which the plausible tracks disagree, and returns a posterior over tracks and a posterior-weighted prediction for every unmeasured variant. In a pre-registered test on 125 loci from a saturation mutagenesis reporter assay, eight pilot measurements were non-inferior to the matched tracks when these existed (mean +0.010 in Spearman correlation) and improved on the all-track mean by 0.047 when they were hidden ($P = 3 \times 10^{-12}$). With eight random measurements, the same inference was worse than not measuring, and four chosen measurements were as accurate as sixteen random ones. On a second dataset with the cell type hidden, eight pilot measurements beat eight random ones by 0.084. The conclusions held with model checkpoints that had not been trained on the tested loci, and with a second foundation model, Enformer, without changing any parameter. The whole panel can be chosen in a single round. In a third pre-registered test, a reporter assay in induced neurons that no track predicted (matched tracks $\rho = 0.07$), TrackPilot fell 0.01–0.04 below the defaults. A pre-registered applicability check on the same eight measurements detects such assays and falls back to the default, recovering 0.010 there while keeping the gains on the other data.
Availability and implementation: TrackPilot is a Python package and command-line tool (MIT licence) at `https://github.com/ydvislab/AuraGenome-V` [TODO: final repository, PyPI name, Zenodo DOI of the submitted version]. Pre-registrations and all derived results are in the same repository.
Contact: [TODO: email]
Supplementary information: Supplementary data are available at *Bioinformatics* online.

Key words: variant effect prediction, genomic foundation models, active model selection, Bayesian experimental design, massively parallel reporter assays

1. Introduction

Deep sequence-to-function models predict the regulatory consequences of non-coding variants from DNA sequence alone [Zhou and Troyanskaya, 2015, Avsec et al., 2021, Linder et al., 2025]. AlphaGenome, the most recent of them, predicts thousands of genome tracks for hundreds of human cell types and tissues and can score any single-nucleotide variant on all of them [Avsec et al., 2026]. On saturation mutagenesis benchmarks such as the CAGI5 regulatory challenge [Kircher et al., 2019, Shigaki et al., 2019] it is

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among the best available predictors. Its output, however, is not one prediction per variant but about a thousand, one per track.

A laboratory that studies variants in a particular cell type therefore has to choose which tracks to use. The usual choice is the tracks whose biosample name matches the cell line. We found this unreliable. Across the 1,018 CAGE, DNase and ATAC tracks of AlphaGenome, the name-matched track ranked between first and 27th by agreement with measured effects on our development elements, and the choice changed biological conclusions: at the *PKLR* promoter in K562 cells, replacing K562 tracks by HepG2 tracks lowered the fraction of significant variants with the correct sign from 0.96 to 0.69–0.85. Many cell types used in the laboratory, including patient-derived lines, are not profiled, and then there is no name to match. Adapting the model to a new cell type is possible but requires genome-wide profiles of that cell type [Hingerl et al., 2025].

What a laboratory usually can do is measure a few variants, for example in a pilot reporter assay of 8–16 constructs before a larger study. This turns the choice of tracks into a label-efficient model selection problem: decide which variants to measure so that the measurements identify the tracks, or the mixture of tracks, that best predict the rest [Cohn et al., 1996]. Active model selection methods address this for classifiers [Karimi et al., 2021, Okanovic et al., 2025, Kay et al., 2025]. Two features of foundation-model tracks break their assumptions. The tracks are regression outputs on arbitrary scales, so each must be calibrated to the assay with an unknown slope and offset that a few labels cannot pin down. And they are a thousand strongly correlated heads of one network, so a few labels often cannot settle which one is best [Kuboyama, 2026], and averaging over the plausible tracks accounts for this uncertainty [Hoeting et al., 1999]. The problem is not specific to AlphaGenome: any model with many output heads and a user with a few labels faces it.

Here we present TrackPilot, a tool that turns a few measured variants into a decision about which tracks to use. TrackPilot (i) treats each track as a hypothesis with an affine calibration that is integrated out in closed form; (ii) starts from the metadata-matched tracks and moves away from them only when the measurements contradict them; (iii) chooses the variants to measure by the classical criterion for discriminating between models [Box and Hill, 1967]; and (iv) reports a posterior-weighted prediction rather than one selected track. We evaluated it in pre-registered tests on three mutagenesis reporter datasets, including one where it failed, checked the dependence on the model's training exposure, and compared it with simpler panels, few-label regression and active model selection methods.

2. Methods

2.1. Problem

A regulatory element is studied in cell type c. A foundation model gives a predicted effect $x_{t,j}$ for each candidate variant j on each of T tracks (T = 1,018 here). The laboratory can measure the effects y_j of k variants, chosen from a pool of variants that can be built, together with the unmutated reference sequence as a control, so that each effect is measured relative to the reference. We want a prediction $\hat{y}_j$ for every other variant, and the panel S, $|S| = k$, that makes these predictions as accurate as possible. We measure accuracy by the Spearman correlation between $\hat{y}$ and y on variants outside the pool.

2.2. Tracks as calibrated hypotheses

Each track's predictions are scaled over the element but not centred, $z_{t,j} = x_{t,j}/\mathrm{sd}_j(x_{t,\cdot})$, so that the sign keeps the direction of the alternative allele. Hypothesis t states

$$\begin{aligned} &y_j = a + b\, z_{t,j} + \varepsilon_j, \\ &a \sim N(0, \alpha s^2), \quad b \sim N(0, \beta s^2), \quad \varepsilon_j \sim N(0, (1-\beta)s^2), \end{aligned} \tag{1}$$

where s^2 is the mean of y_S^2. Integrating out a and b gives the marginal likelihood of the measured panel,

$$y_S \mid t \sim N\Big(0,\ s^2\left[(1-\beta)I + \alpha \mathbf{1}\mathbf{1}^\top + \beta z_{t,S} z_{t,S}^\top\right]\Big), \tag{2}$$

and the posterior $p(t \mid y_S) \propto p(t)\, p(y_S \mid t)$. The predictive mean $m_{t,j} = E[y_j \mid t, y_S]$ follows from the Gaussian conditional, and TrackPilot reports the Bayesian model average $\hat{y}_j = \sum_t p(t \mid y_S)\, m_{t,j}$ [Hoeting et al., 1999]. The two prior parameters were fixed before any evaluation ($\alpha = 0.1$, $\beta = 0.5$) and never tuned.

2.3. Metadata prior

When the model profiles the cell type, the mean of the metadata-matched tracks is a strong default. TrackPilot adds it as one extra hypothesis that holds prior mass $\pi = 0.95$, with a slope prior centred on a positive value, $b \sim N(\sqrt{\beta}\, s,\ (\beta/4)\, s^2)$, so that a negative slope has prior probability of about 2%. Single tracks share the remaining mass. Tracks whose predictions correlate at $r \geq 0.9$ form a family (average linkage, no labels used), and the prior is split equally between families and then within them, so a family of near-duplicates shares the prior mass of one track. Without matched tracks there is no default and all mass goes to single tracks. π was chosen on the Kircher development elements and frozen before the Siraj test; unlike the pre-registered tests, the rule for this choice was not committed before the development runs.

2.4. Choosing the pilot panel

The panel is built sequentially from the pool. Given the current posterior, the next variant maximises the posterior-weighted variance of the hypotheses' predictive means, $\mathrm{Var}_{t\sim p(t\mid y_S)}[m_{t,j}]$, divided by the expected residual variance. This is the Box–Hill criterion for discriminating between models [Box and Hill, 1967, Rainforth et al., 2024]. Before any label, all predictive means are zero under the symmetric slope prior, so the first step uses each track's standardised prediction. Because the choice depends only weakly on the measured values (Section 3.3), TrackPilot can also return the whole panel before any measurement, by replacing each chosen variant's label with the current posterior-average prediction.

2.5. Applicability check

TrackPilot can only help if some track, or mixture of tracks, follows the assay. After the panel is measured, TrackPilot compares the mixture's log evidence, $\log \sum_t p(t)\, p(y_S \mid t)$, with its value for 200 random permutations of the same labels. If the labels are explained no better than shuffled ones (permutation $P > 0.5$, a threshold chosen on development units), no track carries the assay's signal and TrackPilot returns the label-free default instead.

2.6. Implementation

TrackPilot is a Python package with a command-line interface [TODO: finalise once the package is released; keep in step with the README]. Its input is a table of predicted effects (variants × tracks) with track metadata, so it is not tied to one model;

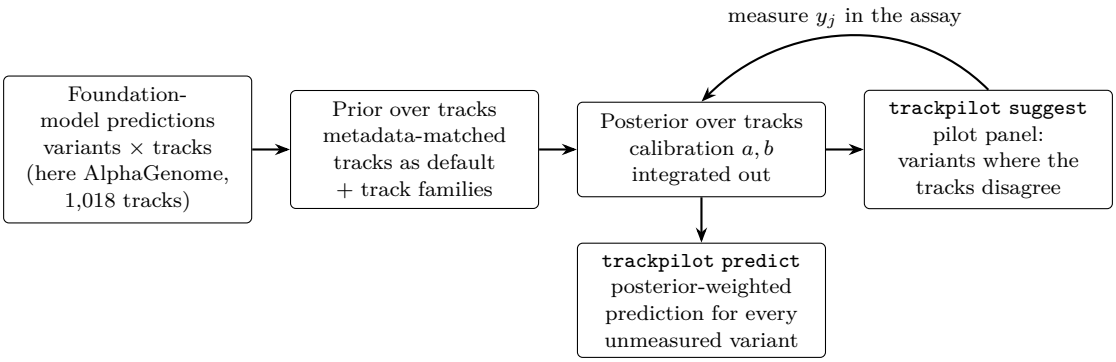


Fig. 1. TrackPilot. Each track is a hypothesis that the measured effect is an affine function of the track's predicted effect. Starting from a prior that favours the metadata-matched tracks, TrackPilot proposes a panel of variants on which the tracks disagree, updates the posterior with their measured effects and predicts every other variant. [TODO: redraw as a two-panel figure: (a) the problem, tracks disagreeing in sign at one locus; (b) the workflow.]

an optional module scores a region with AlphaGenome using the user's own API key (not yet run from the package with a live key). `trackpilot suggest` returns a pilot panel of k variants from a pool; `trackpilot predict` takes the measured effects and returns the posterior over tracks, predictions for all variants, the prediction after the applicability check and a report. For a 1,353-variant element with 1,018 tracks the whole run takes under five seconds on a laptop (Apple M2 Pro). A reproduction script recomputes with the package every frozen TrackPilot result for the first repeat (22 Kircher experiments, 234 numbers; 125 Siraj loci in two cell types, 4,500 numbers) with no difference; the pre-registered Kircher version (uniform prior) is not covered.

2.7. Data and predictions

Kircher et al. [Kircher et al., 2019] measured every possible substitution in 20 disease-associated regulatory elements by saturation mutagenesis MPRA. We used the 22 experiments in human cells (27,789 variants). Five experiments from five regions formed the development set; the other 17 experiments from 11 regions formed the test set. Several test experiments used cell lines AlphaGenome does not profile, such as SK-MEL-28 and glioblastoma lines. **Siraj et al.** [Siraj et al., 2026, 2025] measured all substitutions in 200-bp oligonucleotides around fine-mapped trait variants in K562 and HepG2 cells. We used 125 of the 128 published loci (one excluded for a deletion in the reference, two for having no unique match in GRCh38), 74,975 substitutions each measured in both cell types. The effect is the $\log_2$ fold change of the substituted oligo minus that of the unmutated reference oligo in the same experiment, which a laboratory measuring a pilot panel would also build. **Kosicki et al.** [Kosicki et al., 2025] measured synthetic transversions at every fourth base of 270-bp tiles by lentiviral MPRA in WTC11-derived, NGN2-induced excitatory neurons (ENCODE ENCSR548AQS). We used the 328 forward-strand tiles with at least 30 measured substitutions (22,397 measurements), grouped into 159 regions of overlapping tiles; the effect is the activity of the mutated tile minus that of the reference tile. Neurons are barely profiled: four AlphaGenome and two Enformer tracks match. **Predictions.** We scored every variant by in-silico mutagenesis with AlphaGenome (1-Mb context, default distilled checkpoint) on all CAGE, DNase and ATAC tracks for human biosamples [Avsec et al., 2026], which derive from ENCODE and FANTOM5 [ENCODE Project Consortium et al., 2020, The FANTOM Consortium and the RIKEN PMI and CLST (DGT), 2014]. We scored the neuron regions in the same way, and for Section 3.6 the 11 Kircher test regions, 40 randomly drawn Siraj loci and the neuron regions with Enformer [Avsec et al., 2021] on its 1,322 human CAGE, DNase and ATAC tracks (196-kb context, one forward pass), using the same score: the change in $\log_2$ of the summed signal over a 501-bp window centred on the variant.

2.8. Evaluation design

Scenarios. We hid tracks to simulate three situations: *profiled*, all tracks kept; *related-only*, the element's own cell line hidden; *unprofiled*, the own and related cell types hidden. Related cell types were listed from track metadata before any result was seen. **Protocol.** In each of 200 repeats, half of an element's variants form the pool that could be measured. Every method is scored by Spearman correlation on the same variants outside the pool. The unit of analysis is the region (Kircher; experiments in one region averaged) or the locus (Siraj; two cell types averaged). **Pre-registration.** Two pre-registrations were committed with frozen code before the test data were analysed. For Kircher (version without the metadata prior): H1, the pilot panel beats random panels; H2, it beats the all-track mean by at least 0.03 with a positive difference in at least 7/11 regions. For Siraj (frozen TrackPilot): P1, non-inferiority to the matched mean in the profiled scenario at $k = 8$ (one-sided 95% bootstrap lower bound above -0.02); P2, at most 15% of loci worse by more than 0.05; P3, improvement over the all-track mean when unprofiled at $k = 8$ of at least 0.03; P4, improvement over the related-cell mean when related-only at $k = 16$. A third, for the neurons (N1–N4: the analogues of P3, S1, P1–P2 and P4, all at $k = 8$), was committed before any variant was scored, and a fourth, for the applicability check (A1, A2; Section 3.7), before its test units were run. Analyses added later are labelled post hoc. **Training data, test sets and homology.** TrackPilot trains no model; AlphaGenome is used frozen, and the only labels are the k measured variants of the element being predicted. No measurement from another element and no evaluation variant enters the inference. The two test sets (11 Kircher regions, 125 Siraj loci) were fixed before analysis and do not overlap the development regions. AlphaGenome's training tracks do cover these regions, so we repeated the tests with, for each locus, the checkpoint whose held-out folds contain every training window overlapping the locus (`alphagenome.data.fold_intervals`). These folds are the Borzoi partitions, which assign genome fragments to eight folds at random while keeping human and mouse orthologous regions together [Linder et al., 2025]. Homologous sequence within the human genome is not explicitly grouped, so the held-out-fold results reduce, rather than remove, training exposure. Leave-one-out appears only inside the regression baselines, to choose a penalty. **Baselines.** Label-free: the all-track mean, the matched-track mean

Table 1. Kircher test set, unprofiled scenario (own and related cell types hidden), $k = 8$, 11 regions. Spearman correlation on held-out variants. Pre-registered version (uniform prior); the metadata prior is inactive in this scenario. †Post hoc (first 50 repeats); for LASSO the range covers three rules for all-zero fits.

Panel	Prediction	Mean ρ	TrackPilot − this row	Regions > 0, one-sided P
pilot (TrackPilot)	track posterior	0.370	—	—
random	track posterior	0.286	+0.084	9/11, 0.0024
largest predicted effect	track posterior	0.338	+0.031	–, 0.18
none	all-track mean	0.296	+0.074	8/11, –
random	LASSO†	–	+0.110 to +0.182	9–10/11, ≤0.0024
random	ridge†	–	+0.116	8/11, 0.0068
pilot	LASSO†	–	+0.034 to +0.048	8/11, ≤0.016
pilot	ridge†	–	+0.023	8/11, 0.042
all labels	best single track (reference)	0.454	—	—

and the related-cell mean. Other panels: random, largest predicted effect, and largest unweighted disagreement. Few-label regression: ridge [Hoerl and Kennard, 1970] and LASSO [Tibshirani, 1996] over all tracks; an affine re-fit of the matched mean (`matched_cal`); and a ridge fit on residuals of the calibrated matched mean (`meta_ridge`). Active model selection: CODA- and MODEL SELECTOR-style selection [Kay et al., 2025, Okanovic et al., 2025] with our calibrated likelihood. The best single track chosen with all labels is reported as a reference.

3. Results

3.1. Eight pilot measurements recover a proxy for unprofiled cell types

With the element's own and related cell types hidden, the label-free alternative is the mean of all tracks. On the 11 Kircher test regions, eight pilot measurements raised the mean Spearman correlation from 0.296 (all-track mean) to 0.370, while eight random measurements gave 0.286 (Table 1, Fig. 2). Both pre-registered hypotheses held: the pilot panel beat random panels by 0.084 (9/11 regions, $P = 0.0024$) and the all-track mean by 0.074 (8/11 regions). Gains were largest where the tracks disagreed and the all-track mean was far from the best track, including *IRF4* in SK-MEL-28, a cell line the model does not profile. They were absent in elements with very few significant variants (*FOXE1*, 5).

The pre-registered Siraj test confirmed this on 125 new loci (Table 2, Fig. 3). Without matched or related tracks, eight measurements improved on the all-track mean by 0.047 (0.391 to 0.438; $P = 3 \times 10^{-12}$; P3). With related-cell tracks only, sixteen measurements improved on their mean by 0.040 (0.412 to 0.452; $P = 3 \times 10^{-19}$; P4). Four pilot measurements were as accurate as sixteen random ones (0.402 against 0.404; Supplementary Table S7). The gain was concentrated where the default was poor (post hoc): in the quarter of loci where the all-track mean was furthest from the best track it was 0.133, more than half of that gap, and in the quarter where it was closest, 0.004 (Supplementary Table S8).

3.2. With informative matched tracks, TrackPilot is non-inferior to them

A laboratory working in a profiled cell type would simply use the matched tracks, so the first requirement is not to do worse than them. On Siraj, with all tracks kept, eight measurements were non-inferior to the matched mean (P1: +0.010, one-sided lower bound +0.005 against a margin of −0.02), and only 3/125 loci were worse by more than 0.05 (P2: upper bound 6.1% against 15%). At $k = 16$ the difference was +0.019 with one locus harmed. All four pre-registered hypotheses held. Non-inferiority holds on average, not at every locus: the largest losses were −0.26 (one HepG2 locus) and −0.15 (one K562 locus).

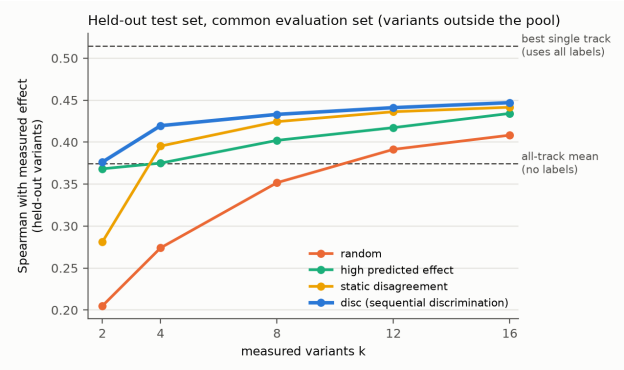


Fig. 2. Kircher test set, unprofiled scenario: Spearman correlation on held-out variants against the number of measured variants k. All panels use the same track posterior. Dashed lines: the label-free all-track mean and the best single track chosen with all labels (a reference).

The metadata prior is what keeps TrackPilot level with the matched tracks. Without it, TrackPilot was 0.029 below the matched mean on the Kircher test regions at $k = 8$, with 7/11 regions harmed (post hoc): it helped where the matched tracks were poor (*LDLR*: matched mean 0.24, TrackPilot 0.51) but lost where they were good (*SORT1*: 0.65 against 0.48). With the prior, chosen on development elements before these runs, it was level with the matched mean (+0.001, one region harmed; +0.018 and none harmed at $k = 16$). The obvious shortcut, re-fitting the matched mean to the same few labels, lowered accuracy on Siraj (−0.082 on random panels, −0.010 on pilot panels; better at only 1/125 loci with either panel), because the fitted slope is poorly determined and can flip sign. On the same panel, TrackPilot was slightly ahead of `meta_ridge` on Siraj (+0.007, 97/125 loci) and tied with it on Kircher.

3.3. Which variants are measured matters; when they are measured does not

Measuring random variants was worse than not measuring: in the unprofiled scenario on Siraj, the same inference reached 0.342 with random panels, below the label-free all-track mean (0.391), and the pilot panel beat random panels at 122/125 loci (S1). The specific criterion mattered less (post hoc; Supplementary Table S4). Choosing variants by the expected information gain about which track is correct matched the pilot panel (differences ≤ 0.003, also with a more accurate approximation; Supplementary Table S6). Panels of the largest predicted effects were as good when matched

Table 2. Pre-registered Siraj test, frozen TrackPilot, 125 loci (two cell types averaged per locus). D is the mean paired difference in Spearman correlation. S rows are secondary analyses.

Hyp.	Comparison	D	Criterion, result
P1	profiled, $k = 8$: TrackPilot − matched mean	+0.010	LB $+0.005 > -0.02$, holds
P2	profiled, $k = 8$: loci with $D < -0.05$	3/125	UB $0.061 \leq 0.15$, holds
P3	unprofiled, $k = 8$: TrackPilot − all-track mean	+0.047	≥ 0.03, $P = 3\times10^{-12}$, holds
P4	related-only, $k = 16$: TrackPilot − related mean	+0.040	> 0, $P = 3\times10^{-19}$, holds
S1	unprofiled, $k = 8$: pilot − random panel	+0.096	$122/125 > 0$
S2	profiled, $k = 8$: TrackPilot − `meta_ridge`	+0.007	$97/125 > 0$
S3	profiled, $k = 8$: `matched_cal` (random panel) − matched	−0.082	$1/125 > 0$

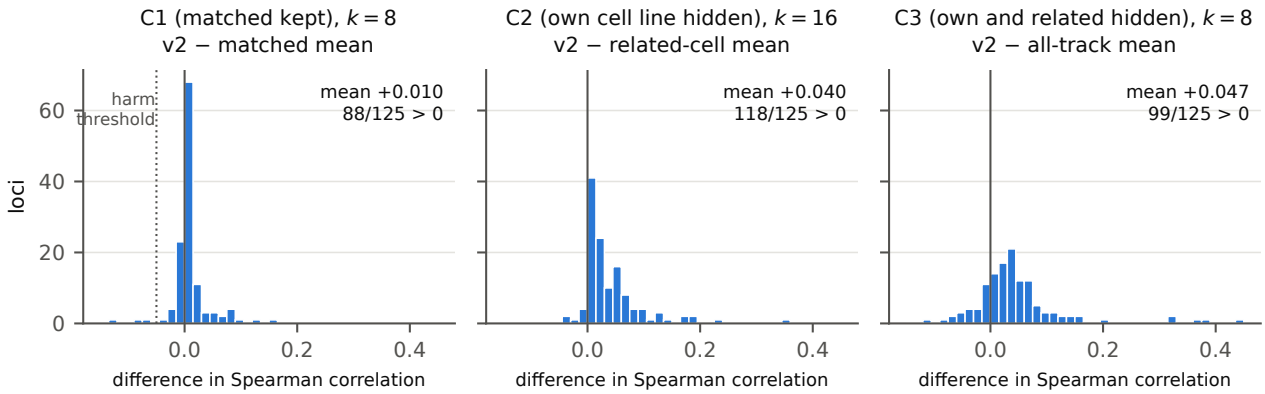


Fig. 3. Per-locus paired differences on the 125 Siraj loci for the pre-registered comparisons. Dotted line: the pre-registered harm threshold (−0.05).

tracks were kept, but worse without them (Siraj unprofiled: −0.028, 97/125 loci). Most useful in practice, a panel chosen in one round, before any measurement, was within 0.005 of sequential choice on both datasets, inside our pre-declared equivalence margin of 0.02. A laboratory can therefore order the whole panel at once. A rule that stops measuring once the prediction stabilises used about four labels with matched tracks and eight without, but lost 0.013–0.027 against sixteen labels.

3.4. Comparison with regression and active model selection

With the same eight labels, regression over all tracks was much worse than TrackPilot (Table 1), and on the pilot panel itself TrackPilot's inference still beat LASSO and ridge by 0.023–0.048. Active model selection methods for classifiers keep a posterior over which candidate is best and report one candidate. We adapted CODA and MODEL SELECTOR [Kay et al., 2025, Okanovic et al., 2025], gave them our calibrated likelihood, and fixed CODA's prior weight on development elements (post hoc; Supplementary Table S5). TrackPilot beat the CODA-style selected track by 0.075 on Siraj with matched tracks kept (111/125 loci), by 0.029–0.034 without them, and by 0.051–0.082 on Kircher. Reporting the baseline's posterior average instead of its single track recovered 0.016–0.026, and with matched tracks hidden the averaged baseline was level with TrackPilot. The advantage therefore comes from two design choices: the metadata prior when matched tracks exist, and averaging over calibrated tracks instead of selecting one.

3.5. Robustness

Training exposure. The default checkpoint was distilled from teachers trained on all training folds, so it has seen tracks over every tested locus [Avsec et al., 2026]. We repeated both tests with, for each locus, the AlphaGenome checkpoint that held out its genomic fold, using reading rules committed before the runs. Absolute accuracies fell by 0.035–0.07, for the matched tracks most (Siraj: 0.474 to 0.402). All four Siraj hypotheses still held, with larger effects (P1 +0.024, P3 +0.050, P4 +0.053; Supplementary Table S3). On the 11 Kircher regions the pilot panel's advantage over random panels grew (+0.093, 10/11, $P = 0.0015$) and its advantage over the all-track mean shrank (+0.054, $P = 0.062$). **Label definition.** Using the published centring of the Siraj effects instead of the reference oligo changed each pre-registered contrast by at most 0.002 (Supplementary Table S1). **Direction of effects.** On Siraj, the fraction of significant held-out variants with the correct predicted sign agreed with the rank results (Supplementary Table S2). On Kircher, the fitted offsets shifted the zero point and TrackPilot called fewer signs correctly than the matched mean (−0.133); without the offsets the deficit shrank to −0.071 (5 regions better, 4 worse; 95% interval −0.21 to +0.01). TrackPilot therefore also reports offset-free predictions for calling directions. **Sharing across loci.** A laboratory often studies several loci in one cell type. In a pre-registered extension, the prior of a new locus mixed the frozen prior with the mean posterior of ten other loci after their own eight-variant panels (weight 0.75, chosen on 40 development loci). On 85 test loci with the cell type unprofiled, this added 0.012 at eight measurements (65/85 loci, $P = 4 \times 10^{-8}$), below the pre-registered 0.02, and 0.022 at four; with matched tracks it stayed non-inferior (+0.014, 3/85 loci harmed; Supplementary Table S9). Sharing helps consistently but modestly.

3.6. A second foundation model

TrackPilot uses only the matrix of predictions and the track metadata, so it should carry over to other models. We re-ran the frozen method, with no parameter changed, on Enformer

Table 3. Frozen TrackPilot on two models. Siraj: 40 random loci, both cell types averaged; Kircher: 11 test regions, unprofiled scenario. Differences in Spearman correlation on held-out variants; regions or loci > 0 and one-sided P in brackets; [a]unprofiled, [b]related cells only. Kircher random-panel P: 0.002 (AlphaGenome), 0.001 (Enformer). Both columns use the frozen method with the family prior; Table 1 shows the pre-registered version.

TrackPilot minus	AlphaGenome	Enformer
Siraj, 40 loci		
matched mean, $k = 8$	+0.006	+0.022
loci worse by > 0.05	1/40	1/40
all-track mean[a], $k = 8$	+0.032 (3e−4)	+0.052 (4e−4)
related mean[b], $k = 16$	+0.036 (4e−10)	+0.047 (5e−7)
matched mean (absolute)	0.473	0.404
Kircher, 11 regions[a], $k = 8$		
random panel	+0.090 (9/11)	+0.097 (10/11)
all-track mean	+0.077 (8/11)	+0.109 (8/11)
TrackPilot (absolute)	0.373	0.324

predictions. The hypotheses and thresholds, the same as in the AlphaGenome tests, were committed before scoring; Siraj was restricted to 40 randomly drawn loci for compute. All six hypotheses held (Table 3). With matched tracks, eight measurements were non-inferior to them (+0.022, one-sided lower bound +0.007; 1/40 loci worse by more than 0.05). Without them, eight measurements improved on the all-track mean by 0.052 on Siraj ($P = 4 \times 10^{-4}$) and by 0.109 on Kircher (8/11 regions), and beat eight random measurements by 0.097 (10/11 regions, $P = 0.001$). Enformer's tracks were less accurate than AlphaGenome's (matched mean 0.404 against 0.473 on the same loci), and TrackPilot's gains were at least as large. Both models' training data cover these regions and include K562 and HepG2, so this test addresses a change of model, not of cell type.

3.7. Detecting assays that no track predicts

On the two main datasets the cell lines are profiled, so the unprofiled scenarios are simulated. As a cell type that the models barely profile, we pre-registered a third test on the neuron assay (N1–N4, both models, no parameter changed). The main hypotheses did not hold (Supplementary Table S10), and the absolute values show why: the matched tracks reached a Spearman correlation of only 0.07 with the measured effects and the all-track mean 0.04–0.05, so the eight labels carried almost no signal that any track shares. TrackPilot was 0.02–0.03 below the matched mean and 0.01–0.04 below the all-track mean, with about 30% of regions worse by more than 0.05.

A user cannot see these absolute values before measuring, but the eight labels themselves show the problem. We therefore pre-registered the applicability check: its threshold was chosen on development units of all three datasets (5 Kircher experiments, 40 Siraj loci, 50 neuron regions) and committed before the held-out units were run. On the 109 held-out neuron regions it judged 43–46% of runs to have no trackable signal, raised accuracy over unchecked TrackPilot by 0.010 (65/109 regions, $P = 0.002$; A1) and stayed within 0.004 of the default. On the 85 held-out Siraj loci it intervened in 5–8% of runs and kept the gains: +0.056 over the all-track mean without matched tracks and +0.011 over the matched mean with them (A2; Supplementary Table S11). The overall neuron results had been seen before the check was designed, so A1 is weaker evidence than the other pre-registered tests.

3.8. Worked example [TODO: pending]

[TODO: Figure: TrackPilot output for one locus, e.g. *IRF4* in SK-MEL-28: (a) the suggested panel on the element; (b) the posterior over track families after 0, 4 and 8 measurements; (c) predicted against measured effects for the held-out variants, compared with the all-track mean. Generated by `trackpilot predict`.]

4. Discussion

TrackPilot gives laboratories a concrete protocol for using a foundation model in their own cell type: start from the tracks whose metadata match, measure about eight variants chosen to split the competing tracks, and let the measurements move the prediction only when they contradict the default. In two pre-registered tests this was non-inferior to the matched tracks when they existed and recovered 0.04–0.05 in Spearman correlation when they were missing or only approximate. On an assay that no track predicted it did not help, and an applicability check on the same measurements identified such cases and returned the default.

Three lessons appear to hold beyond this setting. First, which variants are measured matters as much as how they are modelled; random measurements made the same inference worse than not measuring. Second, re-fitting a default predictor to a handful of labels is harmful, and a prior centred on the default avoids this. Third, the output should be an average: among a thousand correlated heads, the posterior-weighted prediction beat the single most probable track, and at *HBG1* even the best single track chosen with all labels. These lessons should apply to any model with many output heads, because TrackPilot uses only the matrix of predictions and the track metadata; on Enformer all pre-registered hypotheses for the two main datasets held without changing a parameter.

Limitations. K562 and HepG2 are profiled by AlphaGenome, so on Siraj the unprofiled scenarios are simulated by hiding tracks. The one barely profiled cell type we tested in a pre-registered way, induced neurons, came with an assay that the tracks did not predict, so the case of an unprofiled cell type with a predictable assay rests on a few Kircher elements. With eight labels the applicability check has limited power: it recovered only part of the loss on the neuron assay and occasionally discards a useful fit. The evaluation is retrospective. Non-inferiority on average does not exclude losses at single loci (3/125 worse by more than 0.05, the largest −0.26). Absolute correlations are moderate (0.44–0.49 on Siraj), and gains are small where the assay has few significant variants. TrackPilot helps only if at least one track, or a mixture of tracks, is roughly proportional to the assay.

5. Conclusion

A handful of well-chosen measurements is enough to decide which of a thousand foundation-model tracks to use for a cell type, as long as some track carries the assay's signal, the decision starts from the metadata default and the measurements are chosen to discriminate. TrackPilot makes this a two-command step: one to choose the panel and one to turn its measurements into predictions, with a check that tells the user when the measurements follow no track.

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does not accept papers drafted with LLMs; the final text must be written by the authors.]

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Conflict of interest

None declared.

Data availability

Kircher et al. data are available from the original publication [Kircher et al., 2019] [TODO: database accession]. Siraj et al. saturation mutagenesis data are available from Zenodo under CC-BY-NC [Siraj et al., 2025]; they are not redistributed. The neuron MPRA data are ENCODE experiment ENCSR548AQS (files ENCFF958ZLG and ENCFF752NFF, CC BY 4.0) [Kosicki et al., 2025]. AlphaGenome predictions can be regenerated with the public API and the scripts in the repository. Pre-registrations, code and all derived tables are at `https://github.com/ydvislab/AuraGenome-V`.

References

Žiga Avsec, Vikram Agarwal, Daniel Visentin, et al. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021. doi: 10.1038/s41592-021-01252-x.

Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649:1206–1218, 2026. doi: 10.1038/s41586-025-10014-0.

G. E. P. Box and W. J. Hill. Discrimination among mechanistic models. *Technometrics*, 9:57–71, 1967. doi: 10.1080/00401706.1967.10490441.

David A. Cohn, Zoubin Ghahramani, and Michael I. Jordan. Active learning with statistical models. *Journal of Artificial Intelligence Research*, 4:129–145, 1996. doi: 10.1613/jair.295.

ENCODE Project Consortium, Federico Abascal, Reyes Acosta, et al. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*, 583:699–710, 2020. doi: 10.1038/s41586-020-2493-4.

Johannes C. Hingerl, Laura D. Martens, Alexander Karollus, Trevor Manz, Jason D. Buenrostro, Fabian J. Theis, and Julien Gagneur. scooby: modeling multimodal genomic profiles from DNA sequence at single-cell resolution. *Nature Methods*, 22(11): 2275–2285, 2025. doi: 10.1038/s41592-025-02854-5.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1): 55–67, 1970. doi: 10.1080/00401706.1970.10488634.

Jennifer A. Hoeting, David Madigan, Adrian E. Raftery, and Chris T. Volinsky. Bayesian model averaging: a tutorial. *Statistical Science*, 14:382–417, 1999. doi: 10.1214/ss/1009212519.

Mohammad Reza Karimi, Nezihe Merve Gürel, Bojan Karlaš, Johannes Rausch, Ce Zhang, and Andreas Krause. Online active model selection for pre-trained classifiers. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021. arXiv:2010.09818.

Justin Kay, Grant Van Horn, Subhransu Maji, Daniel Sheldon, and Sara Beery. Consensus-driven active model selection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2507.23771.

Martin Kircher, Chenling Xiong, Beth Martin, et al. Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution. *Nature Communications*, 10:3583, 2019. doi: 10.1038/s41467-019-11526-w.

Michael Kosicki, Dianne Laboy Cintrón, Pia Keukeleire, et al. Massively parallel reporter assays and mouse transgenic assays provide correlated and complementary information about neuronal enhancer activity. *Nature Communications*, 16:4786, 2025. doi: 10.1038/s41467-025-60064-1.

Tetsuji Kuboyama. How many labels does model choice need? Certificates and budgets for selective prediction. arXiv:2609.18622, 2026.

Johannes Linder, Divyanshi Srivastava, Han Yuan, Vikram Agarwal, and David R. Kelley. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics*, 57:949–961, 2025. doi: 10.1038/s41588-024-02053-6.

Patrik Okanovic, Andreas Kirsch, Jannes Kasper, Torsten Hoefler, Andreas Krause, and Nezihe Merve Gürel. All models are wrong, some are useful: Model selection with limited labels. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. arXiv:2410.13609.

Tom Rainforth, Adam Foster, Desi R. Ivanova, and Freddie Bickford Smith. Modern Bayesian experimental design. *Statistical Science*, 39:100–114, 2024. doi: 10.1214/23-STS915.

Dustin Shigaki, Orit Adato, Aashish N. Adhikari, Shengcheng Dong, et al. Integration of multiple epigenomic marks improves prediction of variant impact in saturation mutagenesis reporter assay. *Human Mutation*, 40:1280–1291, 2019. doi: 10.1002/humu.23797.

Layla Siraj, Rodrigo Ivan Castro Lopez Vaal, Hannah Dewey, Susan Kales, et al. Functional dissection of complex and molecular trait variants at single nucleotide resolution [dataset]. Zenodo, 2025.

Layla Siraj, Rodrigo I. Castro, Hannah B. Dewey, Susan Kales, et al. Functional dissection of complex trait variants at single-nucleotide resolution. *Nature*, 2026. doi: 10.1038/s41586-026-10121-6.

The FANTOM Consortium and the RIKEN PMI and CLST (DGT). A promoter-level mammalian expression atlas. *Nature*, 507:462–470, 2014. doi: 10.1038/nature13182.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58:267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Jian Zhou and Olga G. Troyanskaya. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods*, 12:931–934, 2015. doi: 10.1038/nmeth.3547.